Equilibrium-Equivalent Reductions to Cone-Leveled Bilinear Games: A Corrected Normalization Theorem and Some Obstructions

Progress note

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Abstract

Dimou asked when a two-player zero-sum game is equilibrium-equivalent to a game with cone-leveled strategy sets and bilinear payoff. This note repairs a common but too-strong reformulation of that question. Positive affine equality of payoffs is not the same as preservation of saddle-point inequalities, and it is much stronger than preservation of equilibria. After making the equivalence notion explicit, we prove: (i) an exact normalization/homogenization theorem for the payoff-representing version of the problem, including a Banach-space variant with closed cones under natural boundedness and continuity hypotheses; (ii) a finite linear-algebra test for biaffine representability on polytopal bases; (iii) a sharp obstruction for finite pure strategy sets under bijective saddle-set equivalence; and (iv) the standard mixed-extension cone-base representation together with the precise criterion under which it introduces no non-pure equilibria.

Status. The note does *not* solve Dimou's full open problem, because that problem does not specify a unique formal meaning of "equilibrium-equivalent." It gives partial progress and a corrected solution of several precise variants. In particular, it solves the global ordinal/payoff-coding variant, gives a complete finite pure-strategy characterization under exact bijective saddle-set equivalence, and identifies the additional hypotheses needed for the normalization theorem to lie in the conic-programming framework.

1 The question and the main correction

A two-player zero-sum game is a triple

$$G = (S, T, u),$$

where player I chooses $s \in S$, player II chooses $t \in T$, and player I receives $u(s,t)$. A pair (s^*, t^*) is a saddle point if

$$u(s, t^*) \le u(s^*, t^*) \le u(s^*, t) \qquad \forall s \in S,\ \forall t \in T. \tag{1}$$

We denote the set of saddle points by $\mathrm{Eq}(G)$.

Dimou's framework studies games whose strategy sets are cone-leveled and whose payoff is bilinear, because such games can be related to conic linear programs. The future-direction question is when a more general zero-sum game is equilibrium-equivalent to one of this form [1]. The open-problem extraction of this question asks for verifiable necessary and sufficient conditions under which the saddle points of G correspond to the saddle points of a game

$$\widetilde{u}(\tilde{x}, \tilde{y}) = \langle \tilde{y}, \widetilde{A}\tilde{x} \rangle$$

with cone-leveled strategy sets [2].

The first point is that the phrase *equilibrium-equivalent* must be specified. There are at least three natural strengths:

Definition 1.1 (Three equivalence notions)**.** *Let $G = (S, T, u)$ and $G' = (S', T', u')$ be two zero-sum games. Assume there are bijections $\phi : S \to S'$ and $\psi : T \to T'$.*

(a) The games are positive-affine payoff equivalent *if there exist $a > 0$ and $c \in \mathbb{R}$ such that*

$$u'(\phi(s), \psi(t)) = au(s, t) + c \qquad \forall (s, t) \in S \times T.$$

(b) The games are globally ordinally payoff equivalent *if there is a strictly increasing function θ on $u(S \times T)$ such that*

$$u'(\phi(s), \psi(t)) = \theta(u(s, t)) \qquad \forall (s, t) \in S \times T.$$

(c) The games are row-column ordinally equivalent *if, for every fixed opponent strategy, the relevant payoff order is preserved and reflected:*

$$u(s_1, t) \le u(s_2, t) \iff u'(\phi(s_1), \psi(t)) \le u'(\phi(s_2), \psi(t)),$$
$$u(s, t_1) \le u(s, t_2) \iff u'(\phi(s), \psi(t_1)) \le u'(\phi(s), \psi(t_2)).$$

(d) The games are saddle-set equivalent *if*

$$(s, t) \in \mathrm{Eq}(G) \iff (\phi(s), \psi(t)) \in \mathrm{Eq}(G').$$

Lemma 1.2 (Implications)**.** *Positive-affine payoff equivalence implies global ordinal payoff equivalence, which implies row-column ordinal equivalence, which implies saddle-set equivalence.*

Proof. The first implication is immediate. A strictly increasing scalar function preserves and reflects all scalar inequalities, hence global ordinal payoff equivalence implies row-column ordinal equivalence. Finally suppose row-column ordinal equivalence holds. If (s^*, t^*) satisfies (1), then the two families of inequalities in (1), one along the column t^* and one along the row s^*, are preserved by the two row-column order assumptions. Thus $(\phi(s^*), \psi(t^*))$ is a saddle point of G'. The converse follows from the reflected implications. □

A positive-affine change of payoff is therefore only one sufficient way to preserve saddle points; it is not "exactly" saddle-inequality preservation.

Example 1.3 (A small counterexample to the affine claim)**.** *Let*

$$M = \begin{pmatrix} 0 & 2 \\ 1 & 3 \end{pmatrix}.$$

For the zero-sum convention in (1)*, the unique saddle point is the lower-left entry, with value* 1*. Replacing every payoff x by e^x preserves every payoff inequality, and therefore preserves the saddle point. But the four numbers $1, e, e^2, e^3$ are not of the form $ax + c$ on $x = 0, 1, 2, 3$: the second finite differences of an affine function vanish, whereas $1 - 2e + e^2 \neq e - 2e^2 + e^3$. Thus saddle-inequality preservation is strictly weaker than positive-affine payoff equivalence.*

The rest of the note records what can be proved once the intended equivalence notion is fixed.

2 Cone-leveled sets and level normalization

We use the following version of Dimou's terminology.

Definition 2.1 (Cone-leveled set)**.** *Let X be a real vector space, let $C \subseteq X$ be a convex cone, let $\alpha : X \to \mathbb{R}$ be a nonzero linear functional, and let $H \subset (0, \infty)$ be nonempty. The set*

$$S = \{x \in C : \alpha(x) \in H\}$$

is called cone-leveled. If $H = \{r\}$ is a singleton, S is basic cone-leveled. When X is a normed space, C is closed, and α belongs to the interior of the positive dual cone C^*, the slice $\{x \in C : \alpha(x) = 1\}$ is called a base of C.

If $S = \{x \in C : \alpha(x) \in H\}$ is cone-leveled, put

$$P = \{p \in C : \alpha(p) = 1\}.$$

Then every $x \in S$ has a unique decomposition

$$x = rp, \qquad r = \alpha(x) \in H, \qquad p = x/\alpha(x) \in P. \tag{2}$$

Conversely, $rp \in S$ for every $(r, p) \in H \times P$. The set P is convex. Thus a cone-leveled set is a set of positive levels multiplying a convex normalized base.

For a bilinear payoff $\widetilde{u}(x, y) = \langle y, Ax\rangle$, this decomposition gives

$$\widetilde{u}(rp, \rho q) = r\rho\, \widetilde{u}(p, q). \tag{3}$$

The restriction $(p, q) \mapsto \widetilde{u}(p, q)$ is affine in each variable. The next definition isolates exactly this normalized structure.

Definition 2.2 (Level-biaffine model)**.** *Let $w : S \times T \to \mathbb{R}$. We say that w admits a level-biaffine model if there are nonempty sets $H_S, H_T \subset (0, \infty)$, convex sets P, Q in real vector spaces, bijections*

$$\Phi : S \to H_S \times P, \qquad \Psi : T \to H_T \times Q,$$

written $\Phi(s) = (r_s, p_s)$ and $\Psi(t) = (\rho_t, q_t)$, and a biaffine function $B : P \times Q \to \mathbb{R}$ such that

$$w(s, t) = r_s \rho_t B(p_s, q_t) \qquad \forall (s, t) \in S \times T. \tag{4}$$

Here biaffine means affine in p for each fixed q and affine in q for each fixed p.

3 The exact payoff-representation theorem

The following theorem is the corrected core of the earlier attempted solution. It is valid, but it solves the payoff-representing variant of the problem rather than the full equilibrium-equivalence problem.

Theorem 3.1 (Algebraic normalization and homogenization)**.** *Let $w : S \times T \to \mathbb{R}$. The following are equivalent.*

(i) There exist real vector spaces X, Y, convex cones $C \subseteq X$, $K \subseteq Y$, linear functionals $\alpha : X \to \mathbb{R}$, $\beta : Y \to \mathbb{R}$, nonempty level sets $H_S, H_T \subset (0,\infty)$, bijections

$$\phi : S \to \{x \in C : \alpha(x) \in H_S\}, \qquad \psi : T \to \{y \in K : \beta(y) \in H_T\},$$

and a bilinear form $\Lambda : X \times Y \to \mathbb{R}$ such that

$$w(s,t) = \Lambda(\phi(s), \psi(t)) \qquad \forall (s,t) \in S \times T.$$

(ii) The payoff w admits a level-biaffine model.

Proof. Assume (i). Let

$$P = \{p \in C : \alpha(p) = 1\}, \qquad Q = \{q \in K : \beta(q) = 1\}.$$

Using (2), write

$$\phi(s) = r_s p_s, \qquad \psi(t) = \rho_t q_t.$$

The maps $s \mapsto (r_s, p_s)$ and $t \mapsto (\rho_t, q_t)$ are bijections onto $H_S \times P$ and $H_T \times Q$, respectively. Define $B(p,q) = \Lambda(p,q)$ on $P \times Q$. Since Λ is bilinear, B is biaffine, and

$$w(s,t) = \Lambda(r_s p_s, \rho_t q_t) = r_s \rho_t B(p_s, q_t).$$

Thus (ii) holds.

Conversely assume (ii). Let $P \subset E$ and $Q \subset F$ be the convex normalized sets in the level-biaffine model. Set

$$X_0 = \operatorname{span}\{(p,1) : p \in P\} \subset E \oplus \mathbb{R}, \qquad Y_0 = \operatorname{span}\{(q,1) : q \in Q\} \subset F \oplus \mathbb{R}.$$

Define cones

$$C = \{\lambda(p,1) : \lambda \geq 0,\ p \in P\} \subset X_0, \qquad K = \{\mu(q,1) : \mu \geq 0,\ q \in Q\} \subset Y_0,$$

and level maps $\alpha(e,r) = r$, $\beta(f,\rho) = \rho$. Convexity of C follows from convexity of P: for $\lambda_1 + \lambda_2 > 0$,

$$\lambda_1(p_1,1) + \lambda_2(p_2,1) = (\lambda_1 + \lambda_2)\left(\frac{\lambda_1 p_1 + \lambda_2 p_2}{\lambda_1 + \lambda_2}, 1\right),$$

and the case $\lambda_1 = \lambda_2 = 0$ is trivial. The same argument applies to K.

We extend B to a bilinear form $\overline{B} : X_0 \times Y_0 \to \mathbb{R}$ by

$$\overline{B}\left(\sum_i a_i(p_i,1), \sum_j b_j(q_j,1)\right) = \sum_{i,j} a_i b_j B(p_i, q_j). \tag{5}$$

This is well-defined. Indeed, suppose $\sum_i a_i(p_i,1) = 0$. Then $\sum_i a_i = 0$ and $\sum_i a_i p_i = 0$. If $f : P \to \mathbb{R}$ is affine, then $\sum_i a_i f(p_i) = 0$: separating the positive and negative coefficients turns the relation into equality of two convex combinations, and affinity gives equality of the corresponding values. Applying this to $f(p) = B(p,q)$ gives $\sum_i a_i B(p_i, q) = 0$ for every $q \in Q$, and then linearly for every element of Y_0. The same argument in the second variable shows that (5) annihilates all linear relations in both arguments. Hence it is a well-defined bilinear form.

Finally put

$$\phi(s) = r_s(p_s,1), \qquad \psi(t) = \rho_t(q_t,1).$$

Then ϕ and ψ are bijections onto the stated cone-leveled sets and

$$\overline{B}(\phi(s), \psi(t)) = r_s \rho_t B(p_s, q_t) = w(s,t).$$

Thus (i) holds. □

The theorem immediately gives the globally ordinal version.

Corollary 3.2 (Global ordinal cone-bilinear reduction)**.** *A game $G = (S, T, u)$ is globally ordinally payoff equivalent to a cone-leveled bilinear game if and only if there exists a strictly increasing function θ on $u(S \times T)$ such that $\theta \circ u$ admits a level-biaffine model.*

Proof. If $\theta \circ u$ has a level-biaffine model, Theorem 3.1 realizes it as a cone-leveled bilinear payoff; strict monotonicity of θ then preserves saddle points by Lemma 1.2. Conversely, any globally ordinal cone-bilinear representation has a numerical payoff $\theta \circ u$ which is cone-leveled bilinear, so Theorem 3.1 applies. $\square$

4 A Banach-space version with closed cones

The algebraic theorem is not automatically a theorem in Dimou's conic-programming setting: one must also control closedness, continuity, duals, and, in the base case, interior positivity of the level functionals. The next proposition gives a clean sufficient set of hypotheses.

Proposition 4.1 (Closed-cone Banach realization)**.** *Let E and F be Banach spaces. Let $P \subset E$ and $Q \subset F$ be nonempty, closed, bounded, convex sets. Let $B : P \times Q \to \mathbb{R}$ be biaffine, and suppose that its algebraic homogenization from* (5) *extends to a continuous bilinear form*

$$\overline{B} : X \times Y \to \mathbb{R},$$

where

$$X = \operatorname{cl}\operatorname{span}\{(p, 1) : p \in P\} \subset E \oplus \mathbb{R}, \qquad Y = \operatorname{cl}\operatorname{span}\{(q, 1) : q \in Q\} \subset F \oplus \mathbb{R}.$$

Then the cones

$$C = \{\lambda(p, 1) : \lambda \geq 0,\ p \in P\} \subset X, \qquad K = \{\mu(q, 1) : \mu \geq 0,\ q \in Q\} \subset Y$$

are closed convex cones. The coordinate functionals $\alpha(e, r) = r$ and $\beta(f, \rho) = \rho$ lie in the interiors of C^* and K^*, respectively. Therefore $\{(p, 1) : p \in P\}$ and $\{(q, 1) : q \in Q\}$ are bases of closed cones. Moreover $\overline{B}$ has the form

$$\overline{B}(x, y) = \langle y, Ax \rangle$$

for a bounded linear operator $A : X \to Y^*$. If E and F are reflexive, then X and Y are reflexive.

Proof. Convexity was proved in Theorem 3.1. We prove closedness of C; the proof for K is identical. Suppose $\lambda_n(p_n, 1) \to (e, r)$ in $E \oplus \mathbb{R}$, with $\lambda_n \geq 0$ and $p_n \in P$. Then $\lambda_n \to r \geq 0$. If $r > 0$, then $p_n \to e/r$, and the closedness of P gives $e/r \in P$, so $(e, r) = r(e/r, 1) \in C$. If $r = 0$, boundedness of P gives $\|\lambda_n p_n\| \to 0$, hence $e = 0$ and $(e, r) = 0 \in C$.

Let $M = \sup_{p \in P} \|p\| < \infty$. For a continuous linear functional $(\ell, a) \in E^* \oplus \mathbb{R}$, positivity on C is equivalent to $\ell(p) + a \geq 0$ for every $p \in P$. If $\|\ell\| < \varepsilon$ and $|a - 1| < \varepsilon$, then

$$\ell(p) + a \geq 1 - \varepsilon - \varepsilon M \qquad \forall p \in P.$$

Choosing $\varepsilon < 1/(M + 1)$ shows that a norm-neighborhood of $\alpha = (0, 1)$ is contained in C^*. Thus $\alpha \in \operatorname{int} C^*$. The proof for β is the same.

Since $\overline{B}$ is continuous bilinear, the map $A : X \to Y^*$ defined by $(Ax)(y) = \overline{B}(x, y)$ is bounded and linear. Finally, if E and F are reflexive, then $E \oplus \mathbb{R}$ and $F \oplus \mathbb{R}$ are reflexive, and closed subspaces of reflexive spaces are reflexive. $\square$

Remark 4.2. *Boundedness in Proposition 4.1 is not cosmetic. Without it, the homogenized cone over a closed set need not be closed. For example, in $\mathbb{R}^2$ let $P = [1, \infty) \subset \mathbb{R}$. Then $C = \{\lambda(p, 1) : \lambda \geq 0,\ p \in P\}$ contains $(1, 1/n)$ for every n but does not contain the limit $(1, 0)$.*

5 A finite test for biaffine payoff tables

For polytopal normalized bases, the payoff-representation condition is a finite linear-algebra check.

Proposition 5.1 (Polytopal biaffine extension test)**.** *Let $P = \operatorname{conv}\{p_1, \ldots, p_m\} \subset E$ and $Q = \operatorname{conv}\{q_1, \ldots, q_n\} \subset F$. A table $M = (M_{ij})$ extends to a biaffine function $B : P \times Q \to \mathbb{R}$ with $B(p_i, q_j) = M_{ij}$ if and only if the following conditions hold:*

$$\sum_{i=1}^{m} a_i = 0, \quad \sum_{i=1}^{m} a_i p_i = 0 \Longrightarrow \sum_{i=1}^{m} a_i M_{ij} = 0 \quad \forall j, \tag{6}$$

$$\sum_{j=1}^{n} b_j = 0, \quad \sum_{j=1}^{n} b_j q_j = 0 \Longrightarrow \sum_{j=1}^{n} b_j M_{ij} = 0 \quad \forall i. \tag{7}$$

When these conditions hold, the extension is

$$B\left(\sum_i \lambda_i p_i, \sum_j \mu_j q_j\right) = \sum_{i,j} \lambda_i \mu_j M_{ij}, \qquad \lambda \in \Delta_m, \quad \mu \in \Delta_n, \tag{8}$$

and the value is independent of the chosen convex representations.

Proof. Necessity follows by applying affine linearity of $B(\cdot, q_j)$ and $B(p_i, \cdot)$ to affine relations among the listed points. Conversely, suppose (6)–(7) hold. If $\sum_i \lambda_i p_i = \sum_i \lambda'_i p_i$ with $\lambda, \lambda' \in \Delta_m$, then $a_i = \lambda_i - \lambda'_i$ is an affine dependence, so (6) implies $\sum_i \lambda_i M_{ij} = \sum_i \lambda'_i M_{ij}$ for every j. Therefore the right side of (8) is independent of the representation of the first argument. The same argument using (7) gives independence of the representation of the second argument. The resulting function is affine in each variable by construction. □

6 Why pure saddle equivalence is much coarser

If equivalence means only equality of saddle sets under a bijection, the payoff away from optimal strategies is mostly invisible.

Lemma 6.1 (Saddle sets are products of optimal-strategy sets)**.** *Suppose $G = (S, T, u)$ has at least one saddle point with value v. Define*

$$S^\star = \{s \in S : u(s,t) \geq v \ \forall t \in T\},$$
$$T^\star = \{t \in T : u(s,t) \leq v \ \forall s \in S\}.$$

Then

$$\operatorname{Eq}(G) = S^\star \times T^\star.$$

Proof. Every saddle point consists of two optimal strategies, hence belongs to $S^\star \times T^\star$. Conversely, if $s \in S^\star$ and $t \in T^\star$, then $u(s,t) \geq v$ by optimality of s and $u(s,t) \leq v$ by optimality of t, so $u(s,t) = v$. For every $s' \in S$ and $t' \in T$,

$$u(s', t) \leq v = u(s,t) = v \leq u(s, t'),$$

so (s,t) is a saddle point. □

This lemma explains why a full answer to Dimou's question cannot be inferred from payoff geometry alone unless the equivalence relation is fixed. The positive-affine and global-ordinal variants preserve much more than $S^\star$ and $T^\star$; pure saddle-set equivalence preserves only these sets.

7 A sharp finite obstruction for exact pure-strategy equivalence

The following result is a useful warning. If the strategy sets themselves are finite and the equivalence is required to be bijective onto finite cone-leveled strategy sets, then cone-leveled bilinear games are extremely restricted.

Lemma 7.1 (Finite cone-leveled sets lie on one ray)**.** *Let*

$$S = \{x \in C : \alpha(x) \in H\}$$

be a finite nonempty cone-leveled set, where C is a convex cone and $H \subset (0, \infty)$. Then there is a point $p_0 \in C$ with $\alpha(p_0) = 1$ such that every $x \in S$ has the form $x = \alpha(x)p_0$. In particular, distinct points of S have distinct levels.

Proof. For $x \in S$, write $\widehat{x} = x/\alpha(x)$, so $\alpha(\widehat{x}) = 1$. Suppose there are $x_1, x_2 \in S$ with $\widehat{x}_1 \neq \widehat{x}_2$. Fix any $h \in H$. For each $\theta \in [0, 1]$ the point

$$z_\theta = h\big(\theta\widehat{x}_1 + (1-\theta)\widehat{x}_2\big)$$

belongs to C, since C is convex and a cone, and satisfies $\alpha(z_\theta) = h$. Thus $z_\theta \in S$ for every $\theta \in [0, 1]$. These points are distinct as θ varies, contradicting finiteness of S. Hence all normalized points are equal to a single p_0. If two points have the same level, they are equal to the same multiple of p_0. □

Theorem 7.2 (Finite exact characterization)**.** *Let $G = (S, T, u)$ be a finite nonempty pure-strategy zero-sum game. Under bijective saddle-set equivalence onto a finite cone-leveled bilinear game, G is representable if and only if its pure saddle set is either a singleton or the whole product $S \times T$.*

Proof. First suppose G' is a finite cone-leveled bilinear game. By Lemma 7.1, after relabeling its strategies we may write

$$x_s = \lambda_s x_0, \qquad y_t = \mu_t y_0,$$

with all $\lambda_s, \mu_t > 0$ and with the λ_s distinct and the μ_t distinct. The payoff is

$$u'(x_s, y_t) = \kappa\lambda_s\mu_t, \qquad \kappa = \Lambda(x_0, y_0).$$

If $\kappa = 0$, every payoff is zero, so every pair is a saddle point. If $\kappa > 0$, player I's best response to every column is the unique strategy with maximal λ_s, and player II's best response to every row is the unique strategy with minimal μ_t. Hence there is exactly one saddle point. If $\kappa < 0$, the same argument gives the unique strategy with minimal λ_s and the unique strategy with maximal μ_t. Thus the saddle set of any finite cone-leveled bilinear game is either a singleton or the whole product.

Conversely, if $\mathrm{Eq}(G) = S \times T$, represent it by the zero bilinear payoff on any finite cone-leveled copies of S and T. If $\mathrm{Eq}(G) = \{(s_0, t_0)\}$, choose distinct positive numbers λ_s with unique maximum at s_0 and distinct positive numbers μ_t with unique minimum at t_0. Let

$$S' = \{\lambda_s : s \in S\} \subset \mathbb{R}_+, \qquad T' = \{\mu_t : t \in T\} \subset \mathbb{R}_+,$$

viewed as cone-leveled subsets of the cone $\mathbb{R}_+$ with level map the identity and with level sets $H_S = S'$, $H_T = T'$. The bilinear payoff $u'(x, y) = xy$ has the unique saddle point corresponding to (s_0, t_0). □

Example 7.3. *The matrix*

$$\begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix}$$

has two pure saddle points, both in the first column. Its saddle set is neither a singleton nor the whole product. Therefore it is not bijectively saddle-set equivalent to any finite cone-leveled bilinear game. Matching pennies has no pure saddle point, so it is also not representable in this exact finite pure-strategy sense.

This theorem is not a negative result about mixed finite games. The usual mixed strategy simplex is already a base of a cone; the obstruction concerns an exact bijection from a finite pure strategy set onto a finite cone-leveled set.

8 Mixed extensions: what they solve and what they do not solve

The canonical way to place an arbitrary payoff table into a cone-base bilinear model is to pass to mixed strategies.

Proposition 8.1 (Finite-support mixed extension). *Let S_0, T_0 be nonempty sets and let $g : S_0 \times T_0 \to \mathbb{R}$ be any payoff. Let $\Delta_f(S_0)$ and $\Delta_f(T_0)$ be the finitely supported probability measures. Then*

$$U(\mu, \nu) = \sum_{s \in S_0} \sum_{t \in T_0} \mu_s \nu_t g(s, t)$$

is a basic cone-leveled bilinear game.

Proof. Let $X = \mathbb{R}^{(S_0)}$ be the vector space of finitely supported signed functions on S_0, let $C = \mathbb{R}_+^{(S_0)}$, and let $\alpha(\mu) = \sum_s \mu_s$. Then

$$\Delta_f(S_0) = \{\mu \in C : \alpha(\mu) = 1\}.$$

Define Y, K, β analogously for T_0. The displayed formula for U is bilinear on $X \times Y$. □

For compact Hausdorff pure strategy spaces and continuous bounded payoff, one may replace finitely supported measures by regular Borel probability measures and the finite sum by the double integral. This gives the familiar mixed-extension cone-base model. It is, however, a representation of the mixed game, not automatically an equilibrium-equivalent representation of the original pure game.

Proposition 8.2 (Criterion for introducing no non-pure equilibria). *Assume a mixed extension has value v. Its optimal mixed strategy sets are*

$$\Delta_S^* = \left\{ \mu \in \Delta_f(S_0) : \sum_s \mu_s g(s, t) \geq v \quad \forall t \in T_0 \right\},$$

$$\Delta_T^* = \left\{ \nu \in \Delta_f(T_0) : \sum_t \nu_t g(s, t) \leq v \quad \forall s \in S_0 \right\}.$$

The equilibrium set of the mixed extension is $\Delta_S^* \times \Delta_T^*$. The mixed extension has only Dirac equilibria, and these are exactly the Dirac images of pure saddle points, if and only if

$$\Delta_S^* = \{\delta_s : s \in S_0,\ \delta_s \in \Delta_S^*\}, \qquad \Delta_T^* = \{\delta_t : t \in T_0,\ \delta_t \in \Delta_T^*\}.$$

Proof. A mixed strategy μ of player I is optimal exactly when it secures at least v against every pure strategy of player II; by linearity this is equivalent to securing at least v against every mixed strategy of player II. This gives the stated formula for Δ_S^*. The formula for Δ_T^* is symmetric. In a valued zero-sum game, the equilibrium set is the product of the two optimal strategy sets. A Dirac pair (δ_s, δ_t) is an equilibrium of the mixed extension exactly when (s, t) is a pure saddle point of the original game. Hence the mixed extension introduces no non-pure equilibria exactly under the displayed Dirac-only condition. □

9 Conclusions and remaining problem

The corrected picture is as follows.

(1) The normalization/homogenization theorem is valid for an explicitly transformed payoff w. Thus the positive-affine version of the previous draft becomes correct when stated merely as one strong payoff-representation variant, and it becomes a slightly broader global-ordinal variant after allowing $w = \theta \circ u$ with θ strictly increasing.

(2) To place the construction in the conic-programming category, extra analytic hypotheses are necessary. Closed bounded normalized bases and a continuous homogenized bilinear form suffice; without boundedness, even closedness of the cone can fail.

(3) Pure saddle-set equivalence is much weaker than payoff equivalence. Under exact bijection of finite pure strategy sets, the cone-leveled bilinear target is so rigid that only games with a unique pure saddle point or with every pair a saddle point can be represented.

(4) Passing to mixed strategies always gives a cone-base bilinear game, but it changes the equilibrium object unless the Dirac-only criterion holds.

Therefore this note is partial progress rather than a full solution to Dimou's future question. A full solution requires first choosing the intended equivalence notion. For payoff or global-ordinal equivalence, Theorem 3.1 and Corollary 3.2 give the exact answer. For mere saddle-set equivalence, the finite characterization above shows that the answer is strongly sensitive to whether one allows enlargement, convexification, or mixed-strategy replacement of the original strategy spaces. The most substantive remaining problem is to characterize row-column ordinal or saddle-set reductions that preserve the strategy spaces in a topologically meaningful way while still producing closed cone-leveled strategy sets and continuous bilinear payoffs suitable for conic duality.

References

[1] Nikos Dimou. *On the Equivalence of Zero-Sum Games and Conic Programs*. Mathematics of Operations Research, published online 2026. DOI: `https://doi.org/10.1287/moor.2025.0926`. arXiv version: `https://arxiv.org/abs/2302.03066`.

[2] Pranav Nuti, Eric Fithian, and Rad Niazadeh. *Open Problems in Operations Research*, problem "Equilibrium-equivalent reduction of zero-sum games to cone-leveled strategy games." Repository page: `https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W7143448186_p0&page=5&n=50`.